SUB-MODULE 7.1 · FINE-TUNING FUNDAMENTALS

Catastrophic Forgetting

Why fine-tuning on new data can erase what the model already knew — and how to stop it

8 min	B3 – Apply	L2 – Intermediate	AI-FT-IN-TH-000004
Duration	Bloom's	Level	ID

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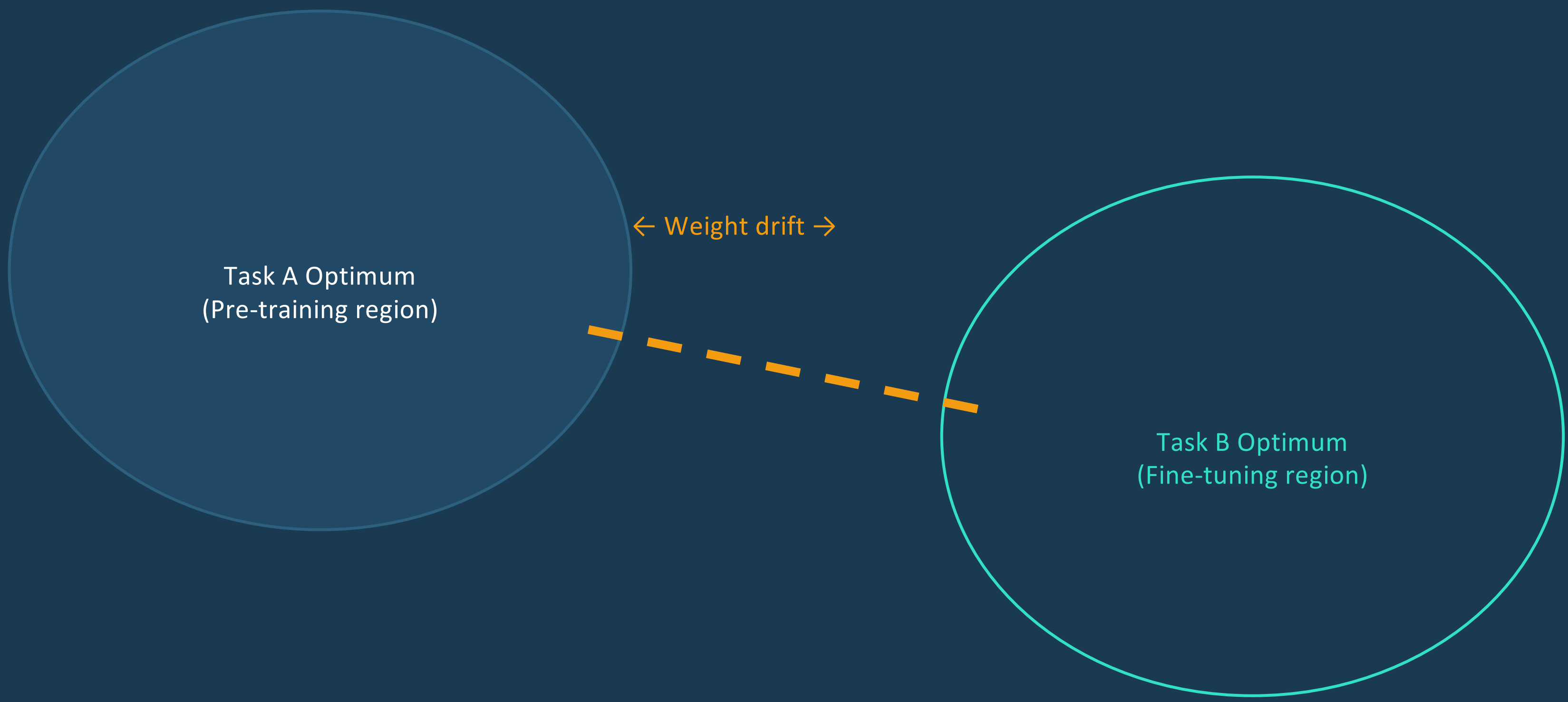
Catastrophic Forgetting — Definition & Mechanism

Definition

Catastrophic forgetting is the phenomenon where a neural network, when trained on a new task, loses the ability to perform tasks it was previously capable of. Gradient updates that minimise loss on the new task move weights away from configurations that supported prior capabilities.

- Also called: catastrophic interference (McCloskey & Cohen, 1989) — a fundamental challenge in continual learning
- Severity scales with distribution shift: fine-tuning medical text on a general model causes moderate forgetting; fine-tuning only on narrow synthetic data causes severe forgetting
- Example: an instruction-following model fine-tuned on SQL generation may lose general instruction-following ability
- The problem is especially acute when the fine-tuning dataset is small — the model overfits and over-writes prior representations

The Mechanism — Weight Space Intuition



Moving weights toward Task B's optimum takes them away from Task A's optimum.
Once Task A performance is lost, it does not recover without re-training on Task A data.

How Severe Is Forgetting? — Empirical Findings

30–60%

Drop in general benchmark performance after narrow domain FT

~10%

Performance loss with EWC regularisation

Indicative

< 3%

Performance loss when using LoRA (PEFT) — base weights frozen

QLoRA

- Extreme: fine-tuning base Llama on only 500 medical Q&A pairs → loses ability to follow general instructions
- Moderate: fine-tuning on 50K domain pairs with data mixing → 10–20% degradation on general tasks
- Minimal: LoRA fine-tuning → only adapter weights change; base weights retain full capability
- Key insight: the smaller and more domain-specific the fine-tuning dataset, the more severe the forgetting

Mitigation Strategy 1 & 2 — Regularisation & Data Mixing

Elastic Weight Consolidation (EWC)

method 1

a variant of Full-finetuning

Add a penalty term to the loss: $L\_EWC = L\_task + \lambda \Sigma_i F_i(\theta_i - \theta^*_i)^2$ Fisher information F_i measures each weight's importance to prior tasks. Heavily penalises changing important weights.

$\theta_i = w_i$

if $= 0 \Rightarrow$ full-fine tuning

$\neq 0 \Rightarrow$ EWC

Data Mixing

Interleave fine-tuning data with a random sample from the original pre-training corpus. If fine-tuning on 10K medical examples, add 2K general examples. Preserves the pre-training distribution signal.

Why EWC Works

Fisher information identifies which directions in weight space matter most for prior tasks. Changes orthogonal to those directions are allowed; changes parallel are penalised.

Why Data Mixing Works

The model sees both old and new tasks simultaneously. Gradient updates must satisfy both objectives, preventing full drift toward the new task optimum.

$\rightarrow$ A $\rightarrow$ K, Q, V, & FFN $\rightarrow$ . . . $\rightarrow$ H $\rightarrow$ P

L

Collection of weights. $\{w_i\}$

We have $\{D^{old}, \& D^{new}\}$

$D^{old} \rightarrow$ [ ] $\rightarrow$ Loss

L

$\frac{\partial L}{\partial w_i}$ $(L = \log(P))$

$\Rightarrow$ how L changes when

$\Rightarrow$ w_i changes

$\Rightarrow$ how w_i is influencing L.

In EWC, update only weights with $\frac{\partial L}{\partial w_i} << 1$

F_i

D^{new}

ie if $\frac{\partial L}{\partial w_i} >> 1 \Rightarrow w_i$ impacts L highly

else $\frac{\partial L}{\partial w_i} << 1 \Rightarrow w_i$ has no impact on L


Dnew
LLm
L

Mitigation Strategy 3 & 4 — Replay & PEFT

Replay Buffers

Store a small set of representative examples from prior tasks. Periodically re-train on this buffer alongside new task data. Used in continual learning systems with sequential task arrival.

PEFT (LoRA / Adapters)

Freeze all base model weights. Train only small adapter layers. Base model capabilities are structurally preserved because its weights are never updated. Near-zero catastrophic forgetting.

When to Use Replay

Sequential task setting — new tasks arrive over time. Budget constraint prevents re-training from scratch. Examples from prior tasks are accessible (not privacy-constrained).

When to Use PEFT

Primary recommendation in most production settings. Negligible forgetting, dramatically lower memory cost, composable adapters for multi-task serving.

Choosing the Right Mitigation Strategy

Scenario	Recommended Strategy	Why
Large compute budget, proprietary dataset	Data mixing	Simple, effective, no architectural change
Sequential task learning (tasks arrive over time)	Replay buffers	Preserves prior task examples
Consumer hardware, limited data	LoRA / QLoRA	Base weights frozen; near-zero forgetting
Academic research on continual learning	EWC	Well-studied; principled regularisation
Production multi-task model	LoRA with multi-adapter serving	Composable; swap adapters at inference

KEY REFERENCES

- McCloskey, M. & Cohen, N.J. (1989). Catastrophic Interference in Connectionist Networks.
- Kirkpatrick, J. et al. (2017). Overcoming Catastrophic Forgetting in Neural Networks. PNAS.
- Hu, E. et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.

▶️ **Next: Concept 5: Transfer Learning Principles (AI-FT-IN-TH-000005)**